

Influence_Consensus_agr_bir - In society, does Agriculture alone generally have a direct effect on the birthrate, or are observed relationships primarily indirect? If a direct effect exists, is it negative, none, negligible, or positive? What would be a reasonable normalized magnitude for this influence on a scale from -1.00 to 1.00? Please only provide evidence for only direct influences.

Agriculture alone can have a direct effect on birthrate, but this influence is generally modest and context-dependent. Agricultural modernization and productivity improvements tend to reduce fertility rates by increasing labor productivity, raising incomes, and facilitating socioeconomic development, which often leads to lower birthrates (Ali, 1985; Gehrke & Kubitz, 2021; Self, 2008). Mechanization and the use of modern farm technologies directly reduce the demand for farm labor, which in turn lowers fertility in farming households by decreasing the economic incentive for larger families (Bhandari & Ghimire, 2013). However, exposure to agricultural pesticides at very high levels has been linked to adverse birth outcomes, indicating a potential negative direct health impact on births rather than fertility rates per se (Larsen et al., 2017). Some studies also highlight that population density in agricultural areas may negatively affect fertility, while crop production itself can have a positive direct effect on birthrate in certain contexts like Bangladesh (Ali, 1985; Chen et al., 2021). Overall, the direct effect of agriculture on birthrate is typically negative or negligible regarding fertility rates, with a reasonable normalized magnitude around -0.2 to 0 reflecting a weak to moderate negative influence primarily through modernization and mechanization rather than agriculture per se (Ali, 1985; Self, 2008; Bhandari & Ghimire, 2013).

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References

- Ali, K. (1985). Determinants of fertility in developing countries.. *Pakistan economic and social review*, 23 1, 65-83.
- Larsen, A., Gaines, S., & Deschenes, O. (2017). Agricultural pesticide use and adverse birth outcomes in the San Joaquin Valley of California. *Nature Communications*, 8. <https://doi.org/10.1038/s41467-017-00349-2>
- Chen, M., Haq, S., Ahmed, K., Hussain, A., & Ahmed, M. (2021). The link between climate change, food security and fertility: The case of Bangladesh. *PLoS ONE*, 16. <https://doi.org/10.1371/journal.pone.0258196>
- Gehrke, E., & Kubitz, C. (2021). Agricultural Productivity and Fertility Rates. *Journal of Human Resources*, 59, 316 - 347. <https://doi.org/10.3368/jhr.0520-10905r1>
- Self, S. (2008). Developing countries and fertility: role of agricultural technology. *International Journal of Development Issues*, 7, 62-75. <https://doi.org/10.1108/14468950810880035>
- Bhandari, P., & Ghimire, D. (2013). Rural Agricultural Change and Fertility Transition in Nepal.. *Rural sociology*, 78 2, 229-252. <https://doi.org/10.1111/ruso.12007>